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second application of JCF

(2) The powers of a matrix are important & can be studied via JCF.

<u>Definition</u> Given $A \in M_n$, we call A

- <u>convergent</u> if $\lim_{m\to\infty} A^m = 0$
- <u>semiconvergent</u> if $\lim_{m\to\infty} A^m = B \in M_n$

where B is a fixed matrix.

$$\left(\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}^m = \begin{bmatrix} 2^m & 0 \\ 0 & 2^m \end{bmatrix} \xrightarrow{m\to\infty} \cancel{\begin{bmatrix} & \\ & \end{bmatrix}} \right)$$

Let's resolve (semi)convergence in the case of a diagonalizable matrix A:

$$A^m \sim \begin{bmatrix} \lambda_1^m & & & 0 \\ & \lambda_2^m & & \\ & & \ddots & \\ 0 & & & \lambda_n^m \end{bmatrix}$$

$$\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$$

So A is convergent iff $|\lambda_j| < 1$, $j = 1, 2, \dots, n$

i.e. $\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\} < 1$

For semiconvergence, we need

$$\lim_{m\to\infty} \lambda_j^m \text{ to exist, } j=1,2,\dots,n$$

So we need $|\lambda_j| \le 1$

If $|\lambda_j|<1$, $\lim_{m\to\infty} \lambda_j = 0$

$|\lambda_j| = 1$, $\lambda_j = 1$

(otherwise $\lambda_j^m \xrightarrow{m\to\infty}$ does not exist)

So A is <u>semiconvergent</u> iff
(diag'ble)

$$\boxed{\begin{array}{c} \rho(A) \le 1 \\ \& \text{ if } |\lambda| = 1 \Rightarrow \lambda = 1 \\ \text{for any } \lambda \in \sigma(A) \end{array}}$$

What about general (non-diagonalizable) matrices?

Say $A \in M_n$, $JCF(A) = J$, i.e.

$$A = SJS^{-1}, \quad J \text{ Jordan matrix}$$

i.e. $A^m = SJ^mS^{-1}$

& $J^m = J_{n_1}^m(\lambda_1) \oplus J_{n_2}^m(\lambda_2) \oplus \dots \oplus J_{n_k}^m(\lambda_k)$

So it is sufficient to study powers (convergence / semiconvergence) of Jordan blocks.

So let's take

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & \lambda \end{bmatrix}$$

$$= \lambda I + \underbrace{\begin{bmatrix} 0 & 1 & & 0 \\ & \ddots & \ddots & \\ & & \ddots & 1 \\ 0 & & & 0 \end{bmatrix}}_{N_k = J_k(0)}$$

Notice

$$N_k^m = 0 \quad \text{for all } m \geq k$$

So

$$J_k^m(\lambda) = (\lambda I + N_k)^m$$

$$= \sum_{i=m-k+1}^{m} \binom{m}{i} \lambda^i N_k^{m-i}$$

So clearly, for $J_k^m(\lambda) \xrightarrow{m \to \infty} 0$, we need

$$|\lambda| < 1 \quad \text{(necessary!)}$$

It is also sufficient because .....

Similarly (homework)

Theorem Let $A \in M_n$. Then A is semiconvergent iff $\rho(A) \leq 1$ and when $|\lambda| = 1$ for some $\lambda \in \sigma(A)$, then

$\rightarrow$ $\lambda = 1$

$\rightarrow$ every Jordan block corresponding to $\lambda = 1$ is 1×1.

To prove this you will need to convince yourself that

if $J_k(\lambda)$, $|\lambda| = 1$

then $\lim_{m \to \infty} J_k^m(\lambda)$ exists iff $\lambda = 1$ & $k = 1$

(e.g., $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^m \xrightarrow{m \to \infty}$ does not exist)

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(3) <u>Theorem</u> Every matrix in M_n can be written as the sum of a diagonalizable matrix and a nilpotent matrix that commute.

<u>Proof</u> Let $A \in M_n$, $J = S^{-1}AS = JCF(A)$.

Notice $J = D + N$, where

$D = \mathrm{diag}(J)$, $N = J - D$.

Notice that N is nilpotent (direct sum of Jordan blocks corresponding to 0)

In fact, if

$$J = J_{n_1}(\lambda_1) \oplus \cdots \oplus J_{n_k}(\lambda_k),$$

then

$$D = \lambda_1 I_{n_1} \oplus \cdots \oplus \lambda_k I_{n_k} \quad : \text{diagonal}$$

$$N = J_{n_1}(0) \oplus \cdots \oplus J_{n_k}(0) \quad :$$

$$N^m = 0, \quad m \geq \max\{n_1, n_2, \ldots, n_k\}$$

nilpotent

Notice also

$$\lambda_j I_{n_j} J_{n_j}(0) = \lambda_j J_{n_j}(0) I_{n_j}$$

$j = 1, 2, \ldots, k$ (easy to verify)

So $DN = ND$, $J = D + N$.

Thus

$$A = SJS^{-1} = \underbrace{SDS^{-1}}_{\text{diagonalizable}} + \underbrace{SNS^{-1}}_{\text{nilpotent}}$$

&

$$\begin{aligned}(SDS^{-1})(SNS^{-1}) &= SD(S^{-1}S)NS^{-1} \\ &= SDNS^{-1} \\ &= SNDS^{-1} \\ &= (SNS^{-1})(SDS^{-1}) \\ &\qquad \text{(commute!)}\end{aligned}$$

Thus A = sum of diag/ble + nilpotent that commute ∎

We can exploit this theorem in computing the exponential of A.

$$\left(\frac{dx}{dt} = Ax(t) \rightsquigarrow x(t) = e^{tA} x(0) \right)$$

Here's how:

<u>Fact</u> $$e^{X+Y} = \sum_{k=0}^{\infty} \frac{(X+Y)^k}{k!} = e^X \cdot e^Y$$

if & only if $XY = YX$.

So now if $A = SJS^{-1}$, $J = JCF(A)$

$J = D + N$ as in previous theorem,

$$D = \begin{bmatrix} \lambda_1 & & & 0 \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix},$$

Then

$$e^A = e^{SJS^{-1}} = \sum_{k=0}^{\infty} \frac{(SJS^{-1})^k}{k!}$$

$$= \sum_{k=0}^{\infty} \frac{SJ^kS^{-1}}{k!}$$

$$= S \sum_{k=0}^{\infty} \frac{J^k}{k!} S^{-1} = Se^J S^{-1}$$

$$= Se^{D+N} S^{-1} \overset{DN=ND}{=} Se^D e^N S^{-1}$$

$$= S \sum_{k=0}^{\infty} \frac{D^k}{k!} \sum_{k=0}^{\infty} \frac{N^k}{k!} S^{-1}$$

$$= S \begin{bmatrix} e^{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & e^{\lambda_m} \end{bmatrix} \sum_{k=0}^{m} \frac{N^k}{k!} S^{-1}, \quad m < \infty$$

computable.

(yet it assumes knowing JCF, which is hard)

Back to the logistics of JCF.

- practice JCF cases
- investigate the "S" ($A = SJS^{-1}$)

<u>Example</u> Let $A \in M_6$ with

$$p_A(t) = (t+3)^4 (t-4)^2$$

Find all possible Jordan Canonical Forms that A can have such that

a) $\operatorname{rank}(A+3I) = 4$

b) $\operatorname{rank}(A+3I)^2 = 2$

c) $\operatorname{rank}(A+3I)^3 = 3$

d) $\operatorname{rank}(A-4I) = 5$ & $\operatorname{rank}(A+3I)^3 = 3$

e) $\operatorname{rank}(A+3I) = 4$, $\operatorname{rank}(A+3I)^2 = 2$ & $\operatorname{rank}(A-4I) = 4$.

a) $\sigma(A) = \{-3, -3, -3, -3, 4, 4\}$

$\sigma(A+3I) = \{0, 0, 0, 0, 7, 7\}$

The Jordan blocks corresponding to 7 (either $J_2(7)$ or $J_1(7) \oplus J_1(7)$) contribute 2 to the rank of $A+3I$.

So the blocks corresponding to 0 must contribute 2 units to the rank of $A+3I$ $(2+2=4)$

So the Jordan blocks corresp. to 0 must be

$$J_2(0) \oplus J_2(0)$$

or

$$J_3(0) \oplus J_1(0)$$

Thus $JCF(A+3I)$

$$= \left\{ \begin{array}{c} J_2(0) \oplus J_2(0) \\ \text{or} \\ J_3(0) \oplus J_1(0) \end{array} \right\} \oplus \left\{ \begin{array}{c} J_1(7) \oplus J_1(7) \\ \text{or} \\ J_2(7) \end{array} \right\}$$

Thus

$$JCF(A) = \left\{ \begin{array}{c} J_2(-3) \oplus J_2(-3) \\ \text{or} \\ J_3(-3) \oplus J_1(-3) \end{array} \right\}$$

$$+ \left\{ \begin{array}{c} J_1(4) \oplus J_1(4) \\ \text{or} \\ J_2(4) \end{array} \right\}$$

4 possible JCF's of A.

(b) Assume now $\operatorname{rank}(A+3I)^2 = 2$

Still Jordan block corresp. to 7 contribute 2 to the rank of $A+3I$ & $(A+3I)^2$

Thus the Jordan blocks corresp. to 0 got annihilating when we found $(A+3I)^2$.

Thus

$$JCF(A+3I) = \left\{ \begin{matrix} J_1(0) \oplus J_1(0) \oplus J_1(0) \oplus J_1(0) \\ J_1(0) \oplus J_1(0) \oplus J_2(0) \\ J_2(0) \oplus J_2(0) \end{matrix} \right\} \oplus \left\{ \begin{matrix} J_1(7) \oplus J_1(7) \\ \text{or} \\ J_2(7) \end{matrix} \right\}$$

$$JCF(A) = \left\{ \begin{matrix} J_1(-3) \oplus J_1(-3) \oplus J_1(-3) \oplus J_1(-3) \\ J_1(-3) \oplus J_1(-3) \oplus J_2(-3) \\ J_2(-3) \oplus J_2(-3) \end{matrix} \right\} \oplus \left\{ \begin{matrix} J_1(4) \oplus J_1(4) \\ \text{or} \\ J_2(4) \end{matrix} \right\}$$

6 possible JCF's of A.